Author: Nothing definite, in the general case. Obviously, one possibility is that (y_n) and (z_n) are convergent. However, it is also possible, for example, for the sequence (y_n) to be convergent, while the sequence (z_n) is divergent.

- Here is a simple illustration:

$$(y_n) = 1, \frac{1}{4}, \frac{1}{9}, \frac{1}{16}, \frac{1}{25}, \ldots\ldots, \frac{1}{n^2}, \ldots\ldots$$
$$(z_n) = 1, 2, 3, 4, 5, \ldots\ldots, n, \ldots\ldots$$
$$(y_n z_n) = 1, \frac{1}{2}, \frac{1}{3}, \frac{1}{4}, \frac{1}{5}, \ldots\ldots, \frac{1}{n}, \ldots\ldots$$

- By the way, note that here we obtain an infinitesimal sequence by multiplying an infinitesimal sequence by an unbounded sequence. In the general case, however, such multiplication need not produce an infinitesimal.

↳ V. Imp. Think about it later

- Finally, there is a possibility when the sequence $(y_n z_n)$ is convergent, and the sequences (y_n) and (z_n) are divergent.

Here is one example: → Ostadharon

$$(y_n) = 1, \frac{1}{4}, 3, \frac{1}{16}, 5, \frac{1}{36}, 7, \ldots\ldots$$
$$(z_n) = 1, 2, \frac{1}{9}, 4, \frac{1}{25}, 6, \frac{1}{49}, \ldots\ldots$$
$$(y_n z_n) = 1, \frac{1}{2}, \frac{1}{3}, \frac{1}{4}, \frac{1}{5}, \frac{1}{6}, \frac{1}{7}, \ldots\ldots$$

Now, let us formulate one more

Theorem: If (y_n) and (z_n) are sequences convergent to a and b when $b \neq 0$, then a sequence $\left(\frac{y_n}{z_n}\right)$ is also convergent, its limit being $\frac{a}{b}$.

We shall omit the proof of this theorem.

Reader: And what if the sequence (z_n) contains zero terms?

Author: Such terms are possible. Nevertheless, the number of such terms can be only finite. Do you know why?

Reader: I think, I can guess. The sequence (z_n) has a nonzero limit b.

Author: Let us specify $b > 0$.